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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Societal Project Synopsis

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Project Title:	**JANANI – A Smart Mobile Companion for Maternal Health, Emergency Hospital Access, and Women's Reproductive Wellness**
Abstract	Maternal and reproductive healthcare in semi-urban and rural India is hindered by limited awareness of nearby healthcare facilities, missed vaccination and check-up schedules, delayed access to hospitals during obstetric emergencies, and low awareness of government welfare schemes and menstrual health conditions such as PCOS/PCOD. This project proposes JANAI, a mobile application that registers a user through her mobile number and hospital details, and offers a chatbot-guided locator for nearby hospitals and Anganwadi centres, SMS-based vaccination and check-up reminders, a month-wise pregnancy tracker with food and health routines, body-change and allergy monitoring, information on maternal and child government schemes, a live emergency location tracker that becomes active in the final trimester to direct users to the nearest hospital, guidance on exercise, diet, postpartum and baby care, and awareness modules on contraception and the menstrual cycle, including PCOS and PCOD. The application is intended to improve continuity of maternal care and reproductive health awareness without offering any medicine identification or dosage suggestions, keeping all clinical decisions with qualified healthcare professionals.
Introduction	Access to consistent, informed maternal and reproductive healthcare remains a challenge for many women, particularly where the nearest hospital, Anganwadi centre or government scheme is not clearly known, and where vaccination schedules and antenatal routines are tracked informally. Smartphone penetration and SMS-based communication offer a low-cost way to close this information gap. JANAI is envisioned as a single companion app that a woman can use from her first pregnancy check-up through delivery, postpartum recovery and beyond, into general reproductive-health stages such as menstrual-cycle tracking and PCOS/PCOD awareness. The application deliberately keeps its chatbot and tracker features limited to locating facilities and services, and sending reminders — it does not identify, recommend, or suggest any medicine, so that every medical decision continues to rest with a doctor or health worker.

Existing System

Paper Details	Observations	Gap Identified	Reference Link
Development of an educational and monitoring mobile application for pregnant women in Nigeria (Babatunde et al., Frontiers in Public Health, 2024)	A five-feature mHealth prototype built with pregnant women at antenatal clinics; raised preeclampsia awareness by 179% and found distance to the nearest facility a major barrier to care.	Provides education and monitoring only; has no live location tracker to the nearest hospital during a high-risk month and no linkage to government maternal schemes.	frontiersin.org/journals/public-health/articles/10.3389/fpubh.2024.1368631
Menstrual Tracking Apps in India: User Perceptions, Attitudes and Implications (Nair, Padmakumar & Krishna, Indian Journal of Marketing)	Survey of 464 Indian women showed good awareness and a positive outlook toward menstrual tracking apps, alongside concerns about data privacy and limited	Cycle tracking stays a standalone feature; it is not linked with pregnancy tracking, hospital/Anganwadi locators, or maternal government schemes.	researcher.manipal.edu (Indian Journal of Marketing, Vol. 53, No. 2)

		medical grounding.			
	Sakhi Swastyay: an inclusive digital platform for menstrual health, PCOD/PCOS awareness (Synapse Social Research Papers)	Combines cycle tracking, an AI conversational assistant, wellness self-assessment and a nearby-facility locator in one platform for reproductive health.	Centred on menstrual/PCOS awareness; does not extend into pregnancy-stage tracking, vaccination-reminder SMS, or postpartum and baby-care guidance.	synapsesocial.com/papers/69a75a9ec6e9836116a20aeb	
Problem Statement	Pregnant women and women managing reproductive health conditions often lack a single, easy-to-use platform that helps them locate nearby hospitals and Anganwadi centres, keeps track of vaccination and check-up schedules, guides them through a month-wise pregnancy routine, alerts them quickly to the nearest hospital during an emergency in the final trimester, and informs them about applicable government schemes, postpartum care, contraception and menstrual health conditions such as PCOS/PCOD — all without depending on informal or unverified sources for medicine-related information.				
Objectives	● To develop a mobile application where users can sign up/sign in with their mobile number and registered hospital details. ● To provide a chatbot-based locator that helps users find nearby hospitals and Anganwadi centres. ● To send automated SMS reminders for scheduled injections/vaccinations and antenatal check-ups to the registered number. ● To build a month-wise pregnancy tracker covering recommended food and health routines. ● To enable body-monitoring for changes and allergies reported by the user. ● To make government schemes for pregnant women, mothers and children easily accessible within the app. ● To provide a live emergency location tracker that activates during the final trimester and directs the user to the nearest hospital for delivery. ● To guide users on exercise, diet, postpartum recovery and baby care. ● To create awareness on contraceptive options and the menstrual cycle, including PCOS and PCOD, for general reproductive-health support.				
Proposed System	JANANI replaces the current fragmented approach — where hospital information, vaccination schedules, government-scheme details and reproductive-health awareness are all sought separately — with one integrated application built around the following modules: ● User module: sign up/sign in using mobile number, along with hospital name registration. ● Locator chatbot: helps the user find nearby hospitals and Anganwadi centres by name or photo-based location search, without any medicine-identification capability. ● Reminder engine: sends SMS reminders for injections/vaccinations and check-ups to the registered number. ● Pregnancy tracker: month-wise routine covering food and health guidance. ● Body-monitoring module: lets the user log body changes and allergies. ● Government schemes module: information on schemes for the mother's and the child's health. ● Emergency locator: a live-location tracker that activates in the ninth month to show the nearest hospital for delivery. ● Wellness module: exercise and diet guidance during and after pregnancy, plus postpartum and baby-care information. ● Reproductive-health module: information on contraception and the menstrual cycle for general women's health, including PCOS and PCOD awareness.				
Methodology	● Requirement gathering and study of existing maternal/reproductive-health mobile applications to finalise the feature set. ● UI/UX design of the sign-up/sign-in flow and each module (chatbot, tracker, reminders, schemes, emergency locator). ● Front-end development of the mobile application. ● Back-end development with a database to store user, hospital/Anganwadi, scheme and reminder data. ● Integration of a maps/location API for the hospital & Anganwadi locator and the emergency live-tracking feature. ● Integration of an SMS gateway API for injection/check-up reminders. ● Development of the chatbot for facility search by name or uploaded photo. ● Testing of each module individually, followed by integration testing of the complete application. ● Deployment and user-acceptance evaluation.				
System Requirement Specifications	**Hardware Requirements** ● Smartphone with Android/iOS (minimum 3 GB RAM) and GPS support ● Stable internet connectivity ● Development machine: minimum i5 processor, 8 GB RAM, 256 GB storage **Software Requirement** ● Android Studio / Flutter or React Native for app development ● Firebase / MySQL for backend database				

	• Google Maps API (or equivalent) for location and emergency tracking • SMS Gateway API for reminders • Figma for UI/UX design
Applications of the Project	• Helps pregnant women quickly locate nearby hospitals and Anganwadi centres. • Reduces missed vaccinations and check-ups through timely SMS reminders. • Supports safer, faster access to hospitals during a delivery-related emergency. • Improves awareness of government maternal and child welfare schemes. • Assists postpartum recovery, baby care and family-planning decisions. • Raises awareness of menstrual health and conditions such as PCOS/PCOD among women in general.
Conclusion	brings together facility location, reminders, pregnancy and postpartum guidance, government-scheme information and reproductive-health awareness into a single application, while intentionally leaving out any medicine-identification or medicine-suggestion feature so that clinical decisions remain with qualified healthcare providers. By doing so, the project aims to improve continuity of maternal care, faster emergency response, and greater awareness of women's reproductive health.
References : (IEEE format)	[1] A. O. Babatunde et al., "Development of an educational and monitoring mobile application for pregnant women in Nigeria," Frontiers in Public Health, vol. 12, 2024. [2] A. Nair, K. Padmakumar, and S. R. Krishna, "Menstrual tracking apps in India: User perceptions, attitudes, and implications," Indian Journal of Marketing, vol. 53, no. 2, pp. 8-25. [3] "Sakhi Swastyay: An inclusive digital platform for menstrual health awareness and early recognition of PCOD/PCOS," Synapse Social Research Papers, 2024. [4] L. M. Daly, D. Horey, P. F. Middleton, F. M. Boyle, and V. Flenady, "The effect of mobile application interventions on influencing healthy maternal behaviour and improving perinatal health outcomes: A systematic review protocol," Systematic Reviews, vol. 6, no. 1, 2017.
Mention the SDG goals that are mapped with this project	• SDG 3 – Good Health and Well-being • SDG 5 – Gender Equality • SDG 10 – Reduced Inequalities

Signature of Guide